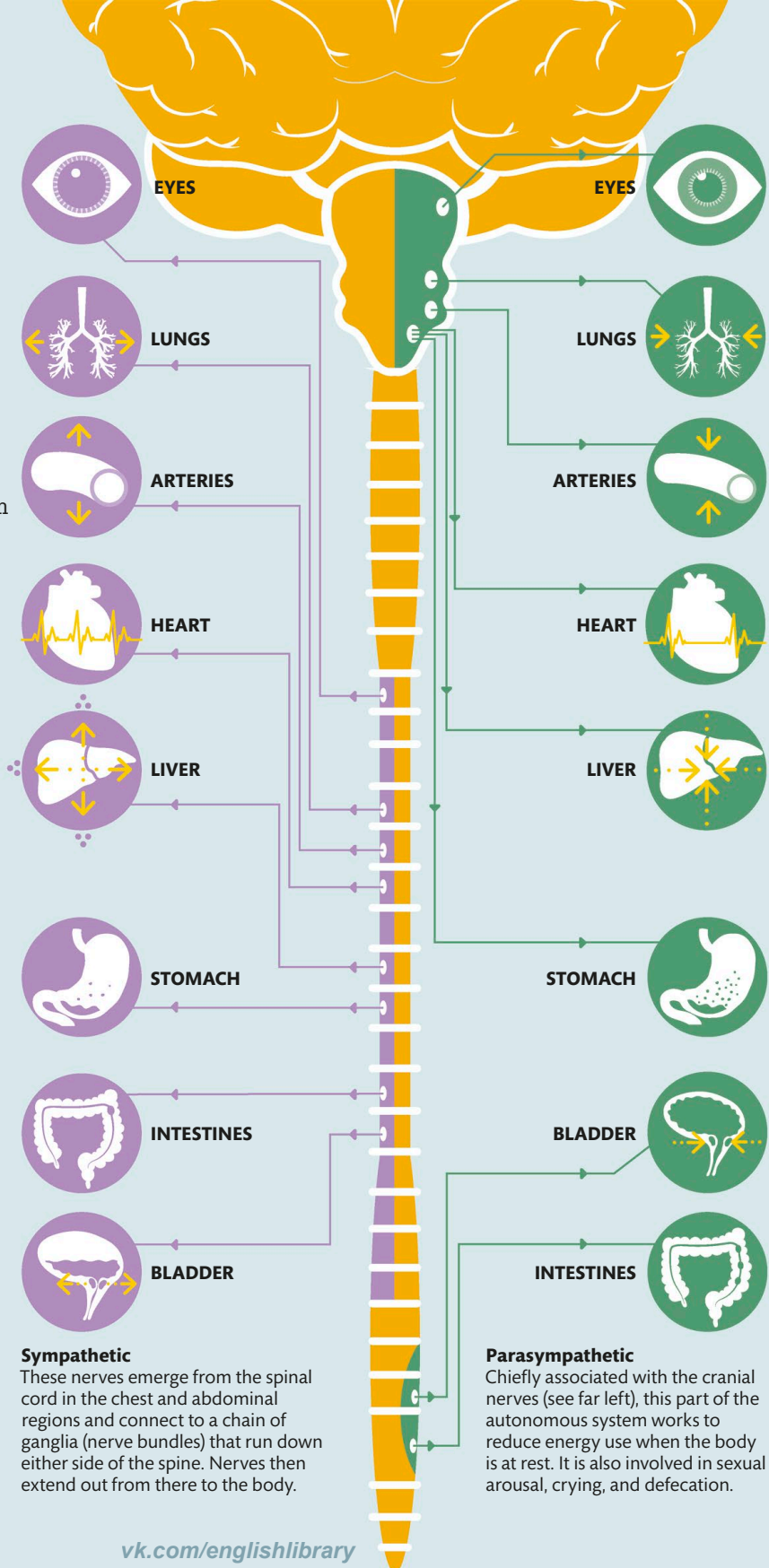




The autonomic nervous system

The involuntary, or autonomic, system maintains the internal conditions of the body by controlling the involuntary muscles in the digestive system and elsewhere, as well as heart and breathing rates, body temperature, and metabolic processes. The autonomic system is divided into two parts. The sympathetic system generally acts to elevate body activity and is involved in the so-called “fight-or-flight” response. The parasympathetic system works in opposition to this, reducing activity to return the body to a “rest-and-digest” state.

THE TOTAL LENGTH
OF THE **SOMATIC**
NERVOUS SYSTEM
IS ABOUT **45 MILES**
(72 KM)



Sympathetic

These nerves emerge from the spinal cord in the chest and abdominal regions and connect to a chain of ganglia (nerve bundles) that run down either side of the spine. Nerves then extend out from there to the body.

Parasympathetic

Chiefly associated with the cranial nerves (see far left), this part of the autonomous system works to reduce energy use when the body is at rest. It is also involved in sexual arousal, crying, and defecation.